College of Computing Education
Bachelor of Science Information Technology

ABC Human Resource Management System
IT6(8419) Fundamentals of Database Systems

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I. Introduction

The proposed web application is a Human Resource Management System (HRMS) developed using the Laravel framework. This system is designed to streamline and automate the management of employee-related processes within an organization. It provides a centralized platform for handling essential HR functions such as employee records, attendance tracking, payroll management, recruitment, and performance evaluation.

Many organizations still rely on manual processes or outdated systems to manage human resources, which often leads to inefficiencies, data inaccuracies, and time-consuming administrative tasks. This HRMS aims to address these issues by offering a modern, web-based solution that improves efficiency, accuracy, and accessibility of HR data.

II. Problem Statement

1. Employee information is stored in paper files or spreadsheets, making it difficult to access and manage.
2. Manual attendance tracking leads to errors and inaccurate records.
3. Payroll processing is time-consuming and prone to miscalculations.
4. Leave requests are handled manually, causing delays and poor tracking.
5. Lack of a centralized system results in data duplication and inconsistency.
6. Generating reports and monitoring employee performance is difficult.
7. Data security is weak due to manual handling of sensitive information.

III. Proposed Solution

1. Implement a centralized HRMS to store and manage employee information efficiently.

2. Automate attendance tracking to improve accuracy and reduce errors.
3. Develop a payroll system that automatically calculates salaries.
4. Provide an online leave management system for faster request processing.
5. Integrate all HR functions into one system to avoid data duplication.
6. Generate real-time reports for better monitoring and decision-making.
7. Enhance data security through user authentication and role-based access.

IV. Objective of the Study

The general objective of this study is to design and develop a Human Resource Management System (HRMS) that automates and streamlines human resource processes such as employee management, attendance tracking, payroll processing, and leave management to improve efficiency, accuracy, and data security within the organization.

1. To develop a centralized database for storing and managing employee information securely.
2. To automate attendance tracking to reduce manual recording errors and improve accuracy.
3. To design a payroll management module that calculates salaries efficiently and accurately.
4. To implement an online leave management system for faster processing of employee leave requests.
5. To provide an easy-to-use interface for both HR personnel and employees.
6. To generate real-time reports that support better decision-making and monitoring of HR activities.
7. To improve data security and reduce redundancy in HR records.

V. Scope and Limitations

Human Resource Management System (HRMS) revolves around automating and controlling the various tasks performed in Human Resource Management. HRMS has a number of functional areas that can be used for employee information management, attendance management, payroll, leaves management, and basic reports generation. The central database will serve as a tool which can be easily accessed, managed and updated by HR professionals who can use this data for better decision-making in terms of recruitment, performance management, etc.

There are several limitations associated with the system. First of all, the system cannot be considered a complete solution due to the fact that some advanced HRM features such as recruitment management or performance analysis using AI-based analytics will have to be developed separately from this software. In addition, the system cannot be used outside the organization, meaning that the users have to maintain internet access to the server at any time they need it.

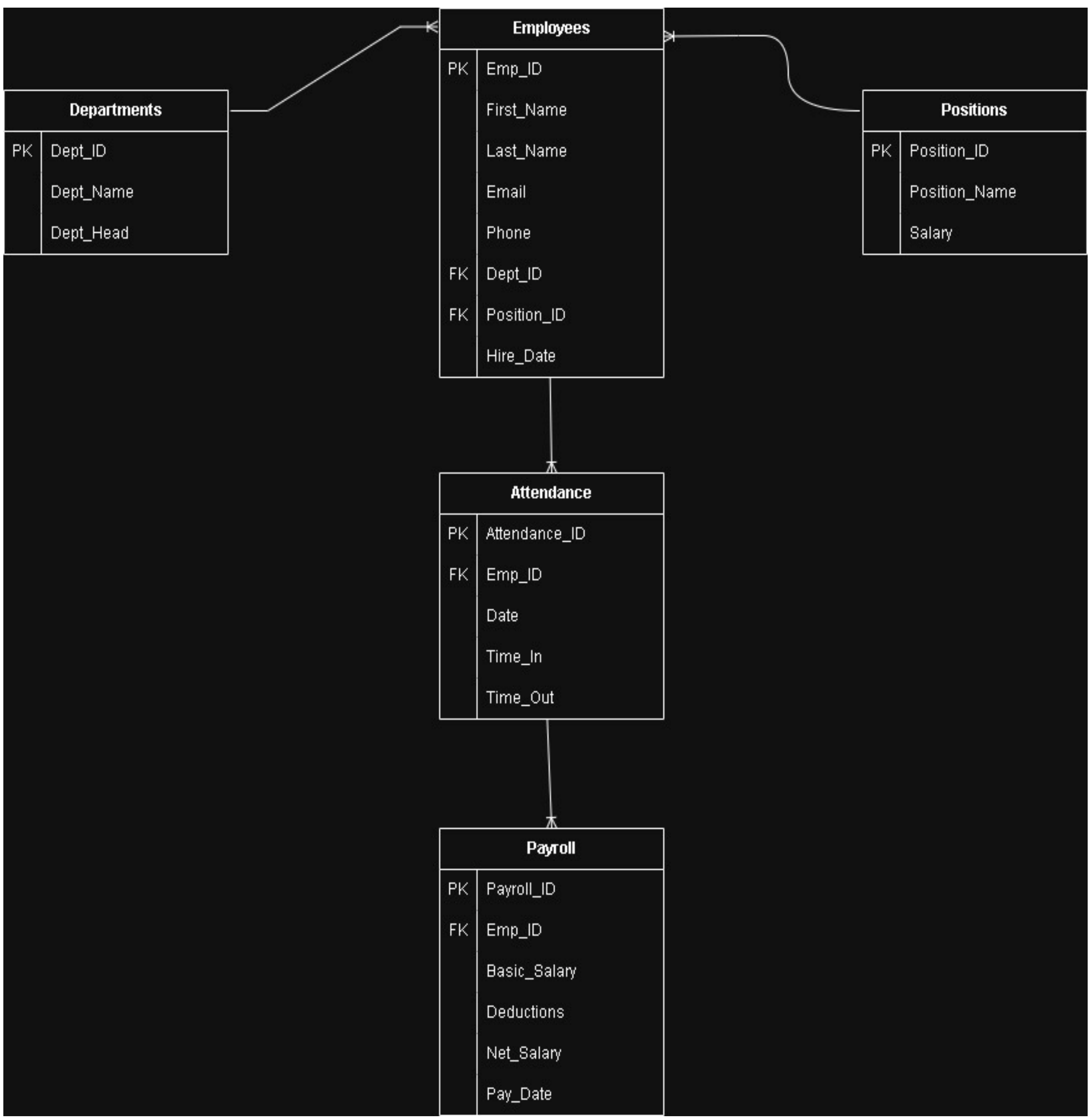
VI. System Analysis and Design

a. Business Rules

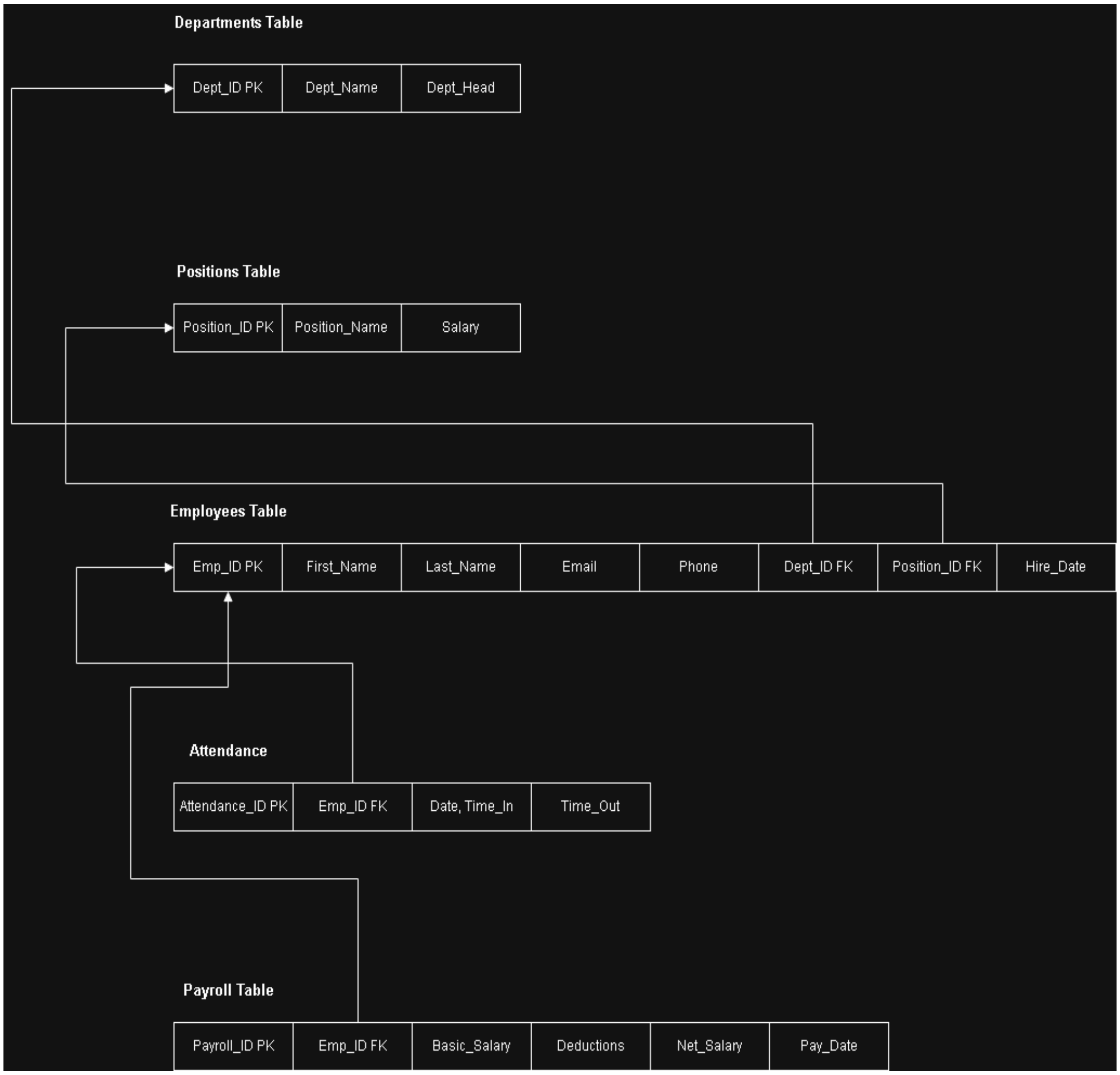
1. Each employee must have a unique ID, and their information must be recorded before accessing the system.
2. Only authorized users (admin/HR) can add, update, or delete employee records.
3. Employees are allowed to view their own information and submit requests only.
4. Attendance must be recorded daily, and any modifications require HR approval.

5. Payroll is calculated based on attendance, approved leaves, and defined salary structure within a set period.
6. Leave requests must be submitted and approved before being applied.
7. The system must prevent duplicate records and ensure all required data fields are completed.
8. User access must be secured through login authentication and role-based permissions.

b. ERD



c. Mapping Entities to Relation



d. Normalization

UNNORMALIZED FORM

emp_id	name	phones	skills	dept_name	position_name	salary	attendance_date	time_in	time_out
101	John Duene	0912, 0933	PHP, Java	IT	Developer	30000	Jan 1, Jan 2	8:00, 8:00 A.M	5:00, 5:00 P.M
102	Felson	945	HTML	HR	Manager	50000	1-Jan	9:00 A.M	6:00 P.M

Table Name: Employee_Record

Fields/Attributes
emp_id, name, phones, skills, dept_name, position_name, salary, attendance_date, time_in, time_out

Primary Key
emp_id

FIRST NORMAL FORM (1NF)

Table Name: Employees

emd_id	name
101	John Duene
102	Felson

Fields/Attributes
emp_id, name

Primary Key
emp_id

Table Name: Employee_Phones

phone_id	emp_id	phone
1	101	912
2	101	933
3	103	945

Fields/Attributes
phone_id, emp_id, phone

Primary Key
phone_id

Foreign Keys
emp_id → Employees(emp_id)

Table Name: Employee_Skills

skill_id	emp_id	skill
1	101	PHP
2	101	Java
3	102	HTML

Fields / Attributes
skill_id, emp_id, skill

Primary Key
skill_id

Foreign Keys
emp_id → Employees(emp_id)

Table Name: Attendance

attendance_id	emp_id	date	time_in	time_out
1	101	1-Jan	8:00	5:00
2	101	2-Jan	8:00	5:00
3	102	1-Jan	9:00	6:00

Fields/Attributes
attendance_id, emp_id, date, time_in, time_out

Primary Key
attendance_id

Foreign Keys
emp_id → Employees(emp_id)

Table Name: Job_Info

emp_id	dept_name	position_name	salary
101	IT	Developer	30000
102	HR	Manager	50000

Fields/Attributes
emp_id, dept_name, position_name, salary

Primary Key
emp_id

Foreign Keys
emp_id → Employees(emp_id)

SECOND NORMAL FORM (2NF)

Table Name: Employees

emp_id	name	dept_id	position_id
101	John Duene	1	1
102	Felson	2	2

Fields/Attributes
emp_id, name, dept_id, position_id

Primary Key
emp_id

Foreign Keys
dept_id → Departments(dept_id)
position_id → Positions(position_id)

Table Name: Departments

dept_id	dept_name
1	IT
2	HR

Fields/Attributes
dept_id, dept_name

Primary Key
dept_id

Table Name: Positions

position_id	position_name	salary
1	Developer	30000
2	Manager	50000

Fields/Attributes
position_id, position_name, salary

Primary Key
position_id

Table Name: Attendance

attendance_id	emp_id	date	time_in	time_out
1	101	1-Jan	8:00	5:00
2	101	2-Jan	8:00	5:00
3	102	1-Jan	9:00	6:00

Fields/Attributes
attendance_id, emp_id, date, time_in, time_out

Primary Key
attendance_id

Foreign Keys
emp_id → Employees(emp_id)

THIRD NORMAL FORM (3NF)

Table Name: Employees

emp_id	name	dept_id	position_id
101	John Duene	1	1
102	Felson	2	2

Fields/Attributes
emp_id, name, dept_id, position_id

Primary Key
emp_id

Foreign Keys
dept_id → Departments(dept_id)
position_id → Positions(position_id)

Table Name: Departments

dept_id	dept_name	location
1	IT	Building A
2	HR	Building B

Fields/Attributes
dept_id, dept_name, location

Primary Key
dept_id

Table Name: Positions

position_id	position_name	salary
1	Developer	30000
2	Manager	50000

Fields/Attributes
position_id, position_name, salary

Primary Key
position_id

Table Name: Payroll

payroll_id	emp_id	basic_salary	deductions
1	101	30000	2000
2	102	50000	5000

Fields/Attributes
payroll_id, emp_id, basic_salary, deductions

Primary Key
payroll_id

Foreign Keys
emp_id → Employees(emp_id)

VII. Database Design

a. List of Tables

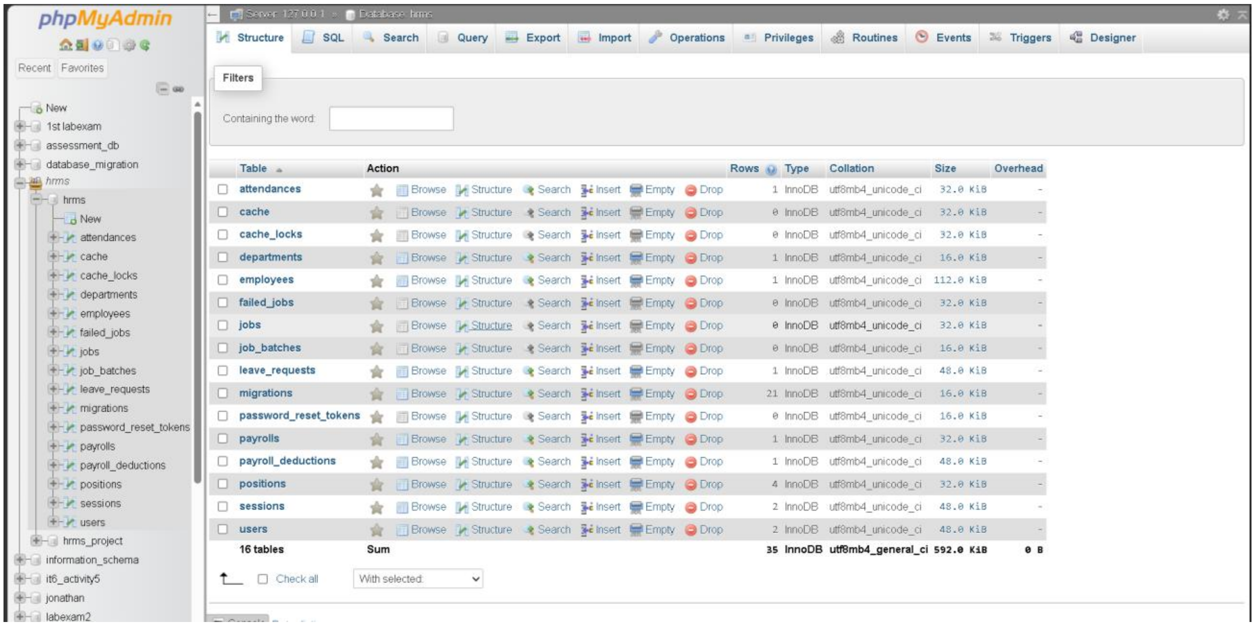


Table	Action	Rows	Type	Collation	Size	Overhead
attendances	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
cache	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
cache_locks	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
departments	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
employees	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	112.0 KiB	-
failed_jobs	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
jobs	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
job_batches	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
leave_requests	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
migrations	Browse Structure Search Insert Empty Drop	21	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
password_reset_tokens	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
payrolls	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
payroll_deductions	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
positions	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
sessions	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
users	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
16 tables	Sum	35	InnoDB	utf8mb4_general_ci	592.0 KiB	0 B

b. Fields and Data Types

Attendance Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**attendance_id**	bigint(20)		UNSIGNED	No	*None*		AUTO_INCREMENT	Change Drop More
2	**emp_id**	bigint(20)		UNSIGNED	No	*None*			Change Drop More
3	**date**	date			No	*None*			Change Drop More
4	**time_in**	time			No	*None*			Change Drop More
5	**time_out**	time			Yes	*NULL*			Change Drop More
6	**status**	enum('present', 'late', 'absent', 'excused')	utf8mb4_unicode_ci		Yes	*NULL*			Change Drop More
7	**late_minutes**	int(11)			No	0			Change Drop More
8	**overtime_hours**	decimal(5,2)			No	0.00			Change Drop More
9	**overtime_type**	enum('regular', 'rest_day', 'holiday')	utf8mb4_unicode_ci		Yes	*NULL*			Change Drop More
10	**has_excuse**	tinyint(1)			No	0			Change Drop More
11	**created_at**	timestamp			Yes	*NULL*			Change Drop More
12	**updated_at**	timestamp			Yes	*NULL*			Change Drop More

Departments Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**dept_id**	bigint(20)		UNSIGNED	No	*None*		AUTO_INCREMENT	Change Drop More
2	**dept_name**	varchar(255)	utf8mb4_unicode_ci		No	*None*			Change Drop More
3	**dept_head**	varchar(255)	utf8mb4_unicode_ci		Yes	*NULL*			Change Drop More
4	**created_at**	timestamp			Yes	*NULL*			Change Drop More
5	**updated_at**	timestamp			Yes	*NULL*			Change Drop More

Employees Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**emp_id**	bigint(20)		UNSIGNED	No	*None*		AUTO_INCREMENT	Change Drop More
2	**user_id**	bigint(20)		UNSIGNED	Yes	*NULL*			Change Drop More
3	**first_name**	varchar(255)	utf8mb4_unicode_ci		No	*None*			Change Drop More
4	**last_name**	varchar(255)	utf8mb4_unicode_ci		No	*None*			Change Drop More
5	**email**	varchar(255)	utf8mb4_unicode_ci		No	*None*			Change Drop More
6	**email_hash**	varchar(64)	utf8mb4_unicode_ci		Yes	*NULL*			Change Drop More
7	**phone**	varchar(255)	utf8mb4_unicode_ci		No	*None*			Change Drop More
8	**dept_id**	bigint(20)		UNSIGNED	No	*None*			Change Drop More
9	**position_id**	bigint(20)		UNSIGNED	No	*None*			Change Drop More
10	**basic_salary**	decimal(12,2)			Yes	*NULL*			Change Drop More
11	**daily_rate**	decimal(10,2)			Yes	*NULL*			Change Drop More
12	**hourly_rate**	decimal(10,2)			Yes	*NULL*			Change Drop More
13	**hire_date**	date			No	*None*			Change Drop More
14	**created_at**	timestamp			Yes	*NULL*			Change Drop More
15	**updated_at**	timestamp			Yes	*NULL*			Change Drop More

Positions Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**position_id**	bigint(20)		UNSIGNED	No	*None*		AUTO_INCREMENT	Change Drop More
2	**position_name**	varchar(255)	utf8mb4_unicode_ci		No	*None*			Change Drop More
3	**salary**	decimal(10,2)			No	*None*			Change Drop More
4	**dept_id**	bigint(20)		UNSIGNED	Yes	*NULL*			Change Drop More
5	**created_at**	timestamp			Yes	*NULL*			Change Drop More
6	**updated_at**	timestamp			Yes	*NULL*			Change Drop More

Payroll Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**payroll_id**	bigint(20)		UNSIGNED	No	*None*		AUTO_INCREMENT	Change Drop More
2	**emp_id**	bigint(20)		UNSIGNED	No	*None*			Change Drop More
3	**basic_salary**	decimal(10,2)			No	*None*			Change Drop More
4	**overtime_pay**	decimal(12,2)			Yes	*NULL*			Change Drop More
5	**bonus**	decimal(12,2)			No	0.00			Change Drop More
6	**late_deduction**	decimal(10,2)			No	0.00			Change Drop More
7	**absent_deduction**	decimal(10,2)			No	0.00			Change Drop More
8	**tax_deduction**	decimal(10,2)			No	0.00			Change Drop More
9	**gross_salary**	decimal(12,2)			Yes	*NULL*			Change Drop More
10	**sss_manual**	decimal(12,2)			No	0.00			Change Drop More
11	**philhealth_manual**	decimal(12,2)			No	0.00			Change Drop More
12	**pag_ibig_manual**	decimal(12,2)			No	0.00			Change Drop More
13	**tax_manual**	decimal(12,2)			No	0.00			Change Drop More
14	**total_deductions**	decimal(12,2)			No	0.00			Change Drop More
15	**payroll_month**	varchar(255)	utf8mb4_unicode_ci		Yes	*NULL*			Change Drop More
16	**computation_details**	text	utf8mb4_unicode_ci		Yes	*NULL*			Change Drop More
17	**status**	varchar(255)	utf8mb4_unicode_ci		No	draft			Change Drop More
18	**approved_by**	bigint(20)		UNSIGNED	Yes	*NULL*			Change Drop More
19	**approved_at**	timestamp			Yes	*NULL*			Change Drop More
20	**deductions**	decimal(10,2)			No	0.00			Change Drop More
21	**net_salary**	decimal(12,2)			No	0.00			Change Drop More
22	**pay_date**	date			No	*None*			Change Drop More
23	**created_at**	timestamp			Yes	*NULL*			Change Drop More
24	**updated_at**	timestamp			Yes	*NULL*			Change Drop More

Payroll_Deductions Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**deduction_id**	bigint(20)		UNSIGNED	No	*None*		AUTO_INCREMENT	Change Drop More
2	**payroll_id**	bigint(20)		UNSIGNED	No	*None*			Change Drop More
3	**deduction_type**	varchar(255)	utf8mb4_unicode_ci		No	*None*			Change Drop More
4	**deduction_name**	varchar(255)	utf8mb4_unicode_ci		No	*None*			Change Drop More
5	**amount**	decimal(12,2)			No	*None*			Change Drop More
6	**description**	text	utf8mb4_unicode_ci		Yes	*NULL*			Change Drop More
7	**created_at**	timestamp			Yes	*NULL*			Change Drop More
8	**updated_at**	timestamp			Yes	*NULL*			Change Drop More

Users Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**id**	bigint(20)		UNSIGNED	No	*None*		AUTO_INCREMENT	Change Drop More
2	**name**	varchar(255)	utf8mb4_unicode_ci		No	*None*			Change Drop More
3	**email**	varchar(255)	utf8mb4_unicode_ci		No	*None*			Change Drop More
4	**email_verified_at**	timestamp			Yes	*NULL*			Change Drop More
5	**password**	varchar(255)	utf8mb4_unicode_ci		No	*None*			Change Drop More
6	**role**	enum('admin', 'user')	utf8mb4_unicode_ci		No	user			Change Drop More
7	**remember_token**	varchar(100)	utf8mb4_unicode_ci		Yes	*NULL*			Change Drop More
8	**created_at**	timestamp			Yes	*NULL*			Change Drop More
9	**updated_at**	timestamp			Yes	*NULL*			Change Drop More

c. Foreign Key Constrain

Attendances Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB) Database	Table	Column
Drop	attendances_emp_id_fore ON DELETE CASCADE ON UPDATE RESTRICT	emp_id + Add column	hrms	employees	emp_id

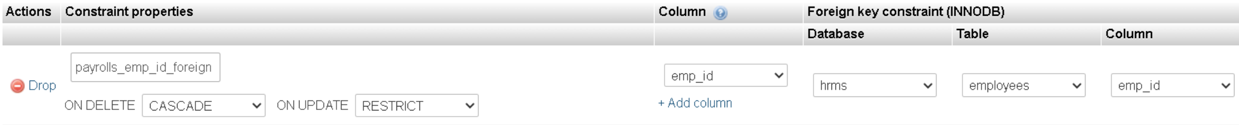
Departments Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB) Database	Table	Column
	Constraint name ON DELETE RESTRICT ON UPDATE RESTRICT	 + Add column	hrms		

Employees Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB) Database	Table	Column
Drop	employees_dept_id_forei ON DELETE RESTRICT ON UPDATE RESTRICT	dept_id + Add column	hrms	departments	dept_id
Drop	employees_position_id_fo ON DELETE RESTRICT ON UPDATE RESTRICT	position_id + Add column	hrms	positions	position_id
Drop	employees_user_id_forei ON DELETE SET NULL ON UPDATE RESTRICT	user_id + Add column	hrms	users	id

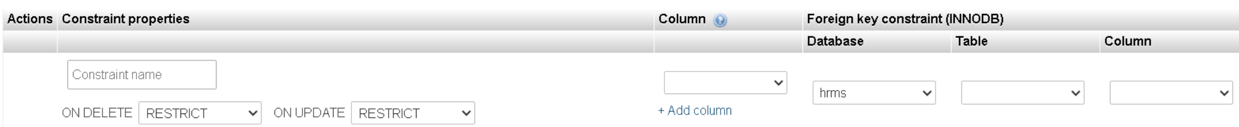
Payrolls Constraint



Payroll_Deductions Constraint

Positions Constraint

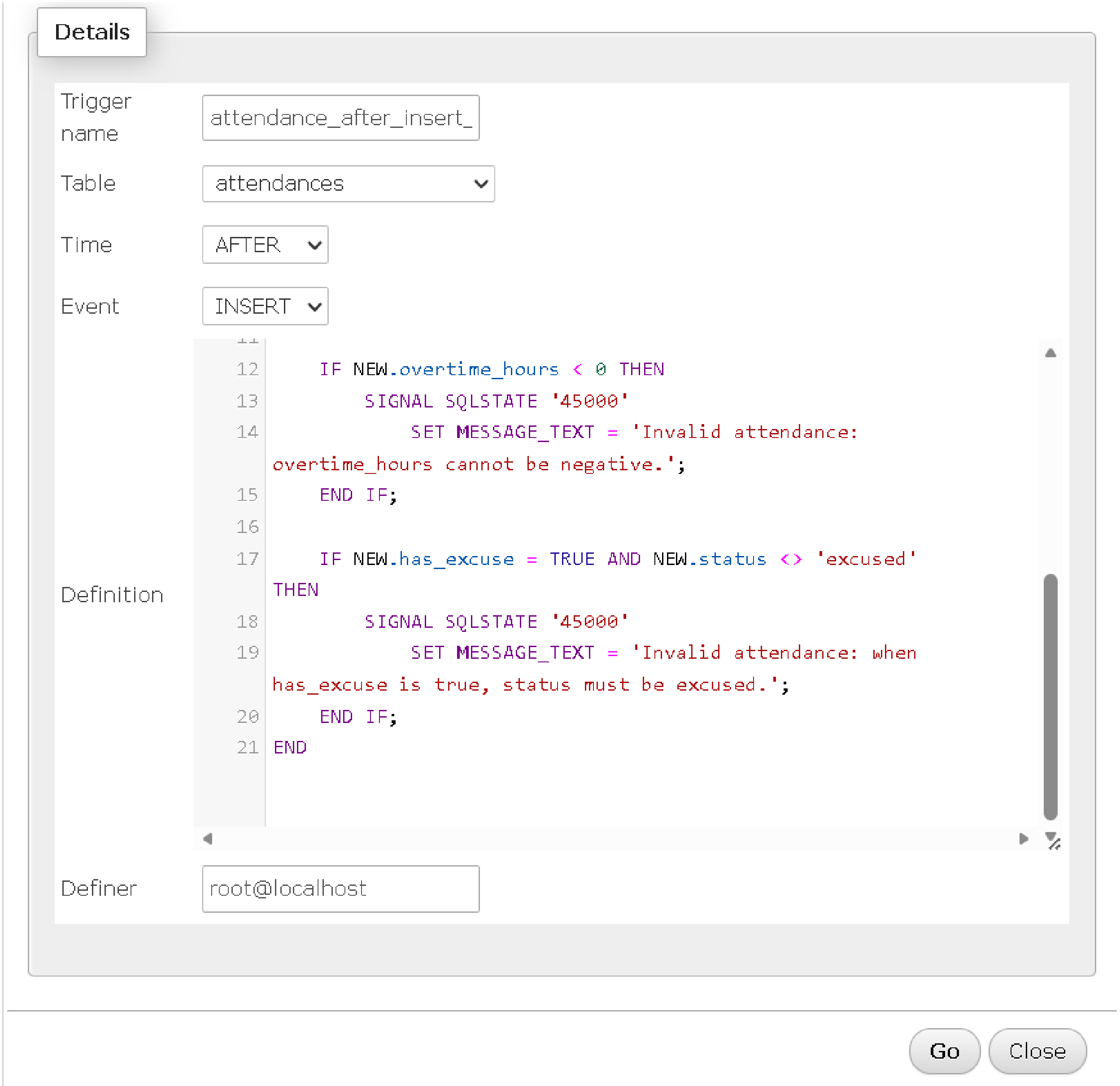
User Constraint



VIII. Advanced Database Features

TRIGGERS

1. attendance_after_insert_validate



Explanation: Trigger BEGIN…END is the block that runs automatically after every INSERT into the attendances table, and each IF statement checks the new row's data (like time_in <= time_out, late_minutes >= 0, and overtime_hours >= 0, and if has_excuse is true then status must be excused); if any rule fails it stops the insert by raising an error using SIGNAL SQLSTATE '45000' with the given message.

2. payroll_after_insert_validate

Details

Trigger name: payroll_after_insert_valic

Table: payrolls

Time: AFTER

Event: INSERT

Definition:

```
BEGIN
    IF NEW.net_salary < 0 THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Invalid payroll:
net_salary cannot be negative.';
    END IF;

    IF NEW.total_deductions < 0 THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Invalid payroll:
total_deductions cannot be negative.';
    END IF;

    IF NEW.basic_salary < 0 THEN
        SIGNAL SQLSTATE '45000'
```

Definer: root@localhost

Go Close

Details

Trigger name: payroll_after_insert_valic

Table: payrolls

Time: AFTER

Event: INSERT

Definition:

```
    END IF;

    IF NEW.gross_salary IS NOT NULL AND NEW.gross_salary < 0 THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Invalid payroll: gross_salary cannot be negative.';
    END IF;

    IF NEW.payroll_month IS NOT NULL AND CHAR_LENGTH(NEW.payroll_month) = 0 THEN
        SIGNAL SQLSTATE '45000'
            SET MESSAGE_TEXT = 'Invalid payroll: payroll_month cannot be empty.';
    END IF;
END
```

Definer: root@localhost

Go Close

Explanation: The BEGIN block checks every new row inserted into payrolls and, if net_salary, total_deductions, or basic_salary are negative (or if gross_salary is provided and negative, or if payroll_month is provided but empty), it stops the insert by raising an error with SIGNAL SQLSTATE '45000' using the message text set in each IF branch.

3. leave_request_after_insert_validate

Explanation: This trigger runs after each new insert into leave_requests and checks that start_date is not after end_date, total_days is not negative, and status (if present) is one of pending, approved, or rejected, otherwise it blocks the insert by raising an error with SIGNAL SQLSTATE '45000' and the provided message.

VIEWS

1. leave_request_details

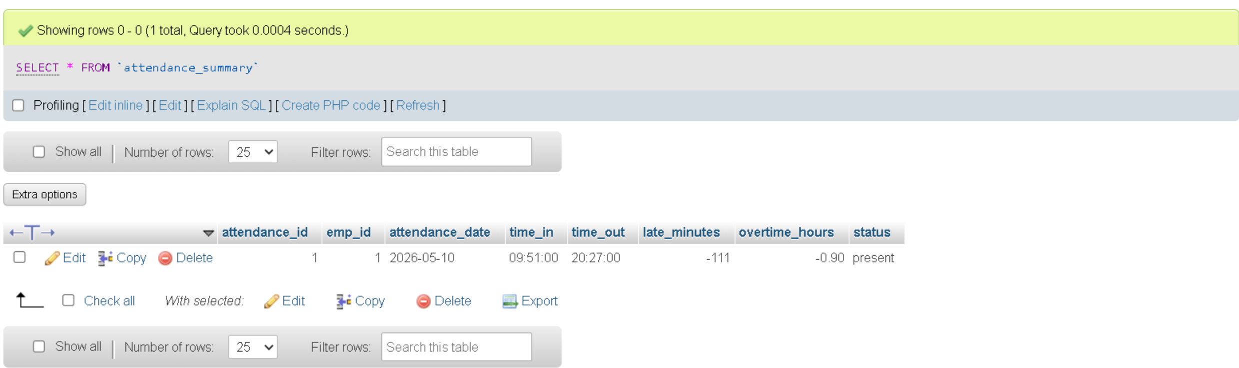
The leave_request_details MySQL VIEW provides a joined/ready-to-query dataset of leave requests with employee details so it appears in phpMyAdmin under Views and can be used for easy reporting/UI queries.

2. payroll_report

The payroll_report MySQL VIEW joins payrolls with employee names so phpMyAdmin can display payroll report data easily for auditing/reporting.

3. attendance_summary

The attendance_summary MySQL VIEW provides a simple dataset of attendance records with a consistent attendance_date column mapping for easier viewing in phpMyAdmin and reporting.



IX. System Features and Functionality

This HRMS system consists of authentication/role-protected user access, and core HR modules for **Departments**, **Positions**, **Employees**, **Attendance**, **Leave Requests**, and **Payroll** each module provides CRUD screens and related views so users can manage master data (departments/positions), maintain employee records, register and validate attendance, submit and track leave requests, and calculate/pay review payroll results, with extra database objects like SQL triggers/views used to support validation and reporting in phpMyAdmin.

1. Login (Admin)
    - Admin also logs in the same way, but then the Admin middleware/role protection restricts access to admin-only pages (such as managing HR master data and workflow screens).

2. Register (Admin)
    - Admin registration follows the same registration mechanism, but in practice admin accounts are expected to have an admin/role set so they pass the AdminMiddleware checks.

3. Departments (Admin)
    - Admin can create, edit, and manage department records.
    - Used to organize employees by organizational unit.

4. Positions (Admin)
    - Admin manages job positions linked to departments.
    - This structure is used when assigning employees to roles/positions.

5. Employees (Admin)
    - Admin creates and maintains employee records (and related HR attributes).
    - Admin also manages updates like department/position association and any other employee fields.

6. Attendance (Admin)
    - Admin records attendance per employee (date, time in/out, status, late/overtime derived data).
    - System uses attendance data as the basis for payroll computations.

7. Leave Requests (Admin)
    - Admin reviews/handles leave request submissions (approve/reject logic where implemented).
    - Helps ensure payroll/attendance impacts are properly reflected.

8. Payroll (Admin)
    - Admin calculates and generates payroll results using attendance and leave outcomes.
    - Includes reporting features using database objects (views) to simplify payroll reporting in tools like phpMyAdmin.

9. Login (User)
    - User logs in using the credentials on the login page.
    - After login, the user is authenticated and shown their dashboard/layout.

10. Register (User)
    - New users register through the register page (name/email/password).
    - After successful registration, the user is typically redirected to email verification/next auth step depending on the configured auth flow.

11. Departments (User)
    - Non-admin users usually cannot edit/create departments; they can view relevant department information if the UI exposes it.

12. Positions (User)
    - Users typically view positions (and related lists) but don’t manage them.

13. Employees (User)
    - Users may view employee details/lists depending on permissions, but editing is restricted to admins.

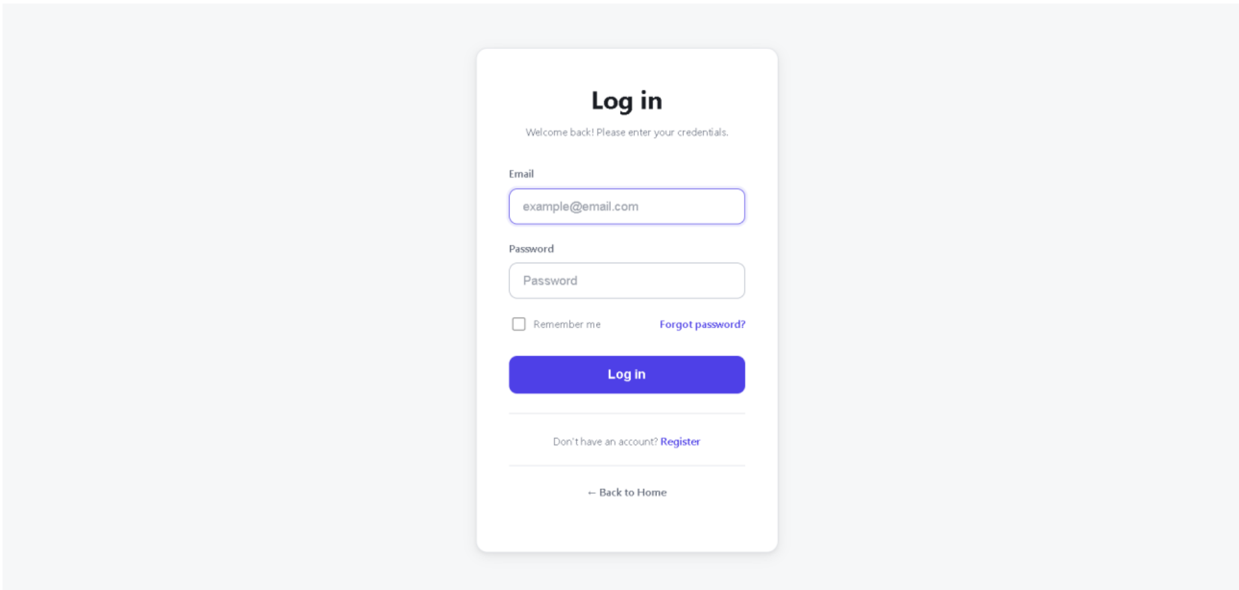
14. Attendance (User)
    - Users typically can view their own attendance (if implemented) but not modify attendance entries.

15. Leave Requests (User)
    - User submits leave requests and can view the status of their own requests.

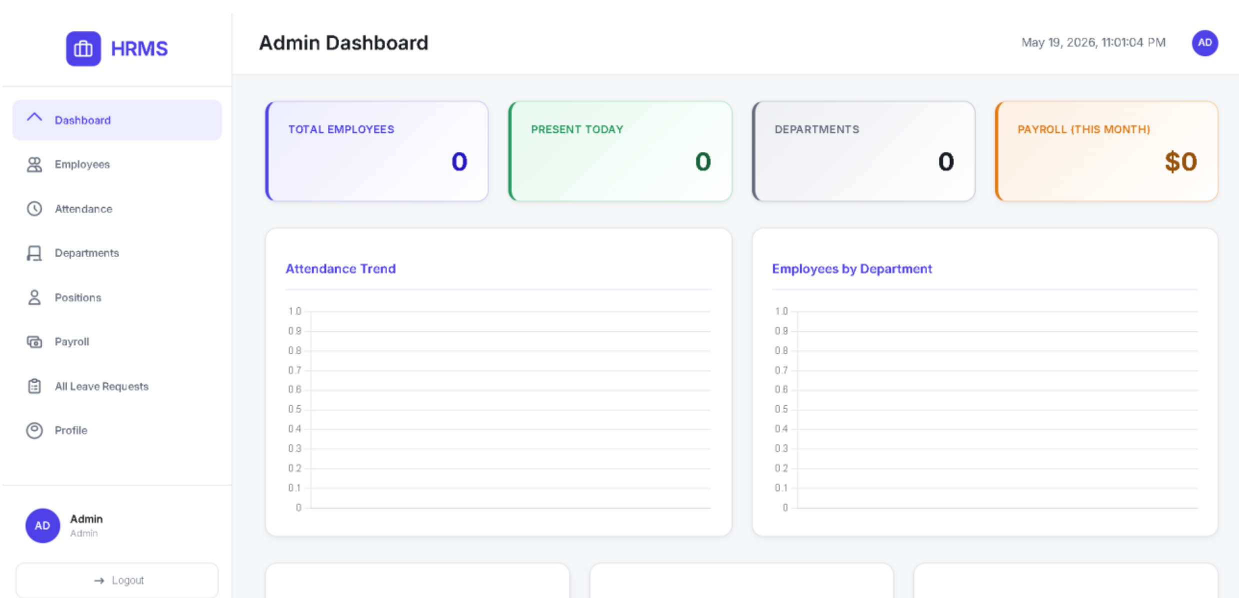
16. Payroll (User)
    - User typically views payroll/pay statements or reports available to them (not generate/adjust payroll).

X. Prototype / User Interface

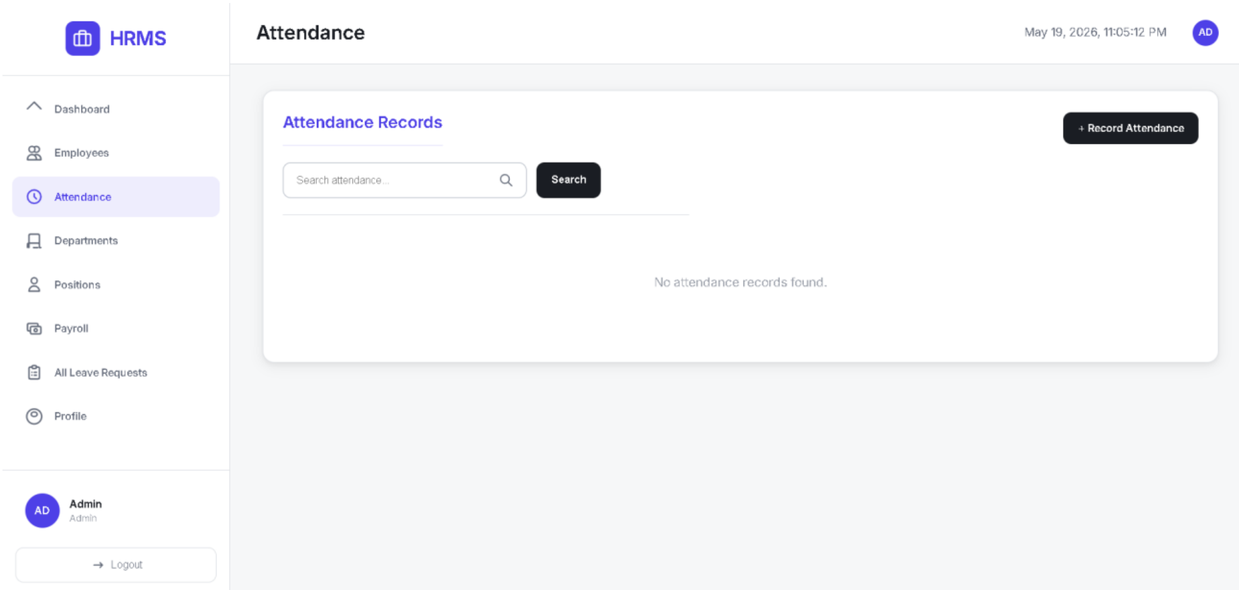
1. Login Page



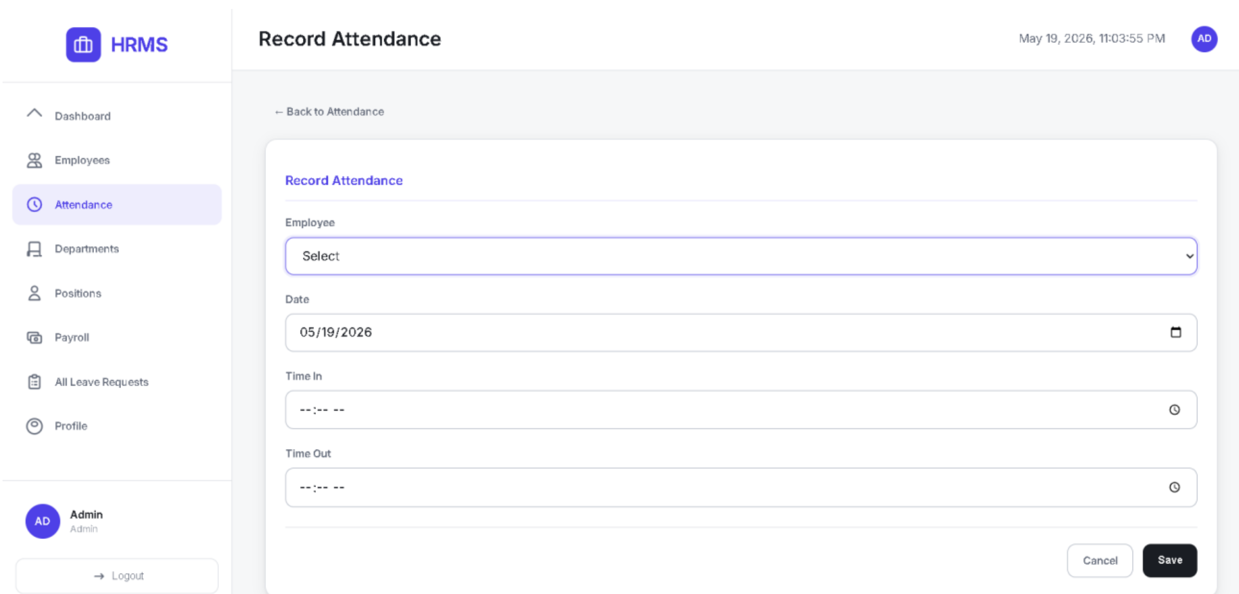
2. Dashboard Page



3. Attendance Records Page (Admin)



4. Attendance Page (Create attendance for Employee)

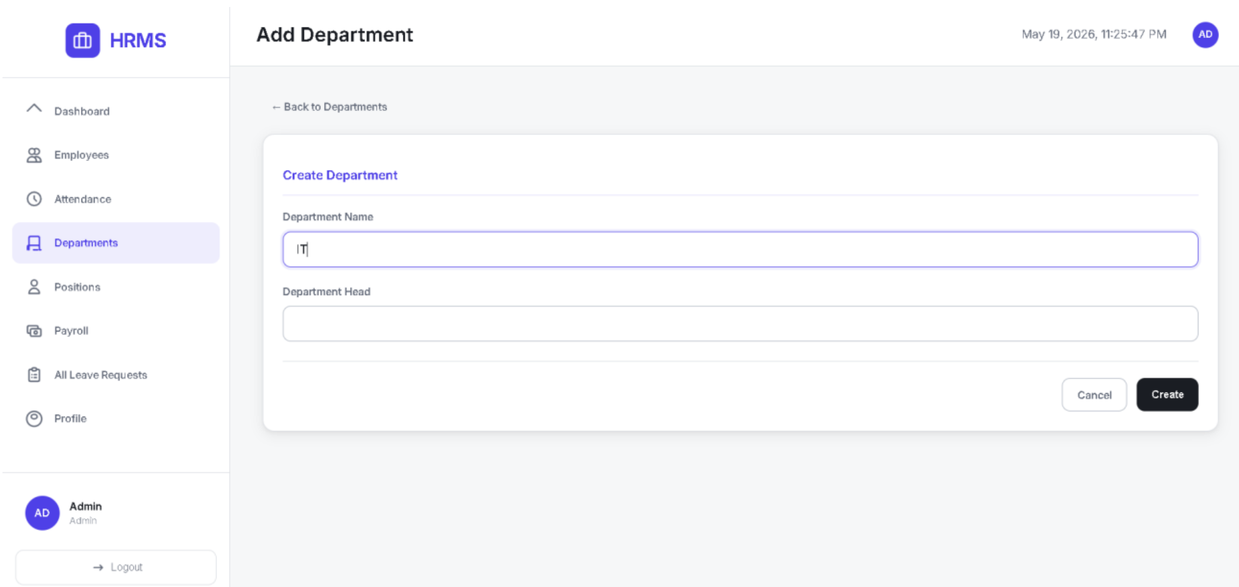


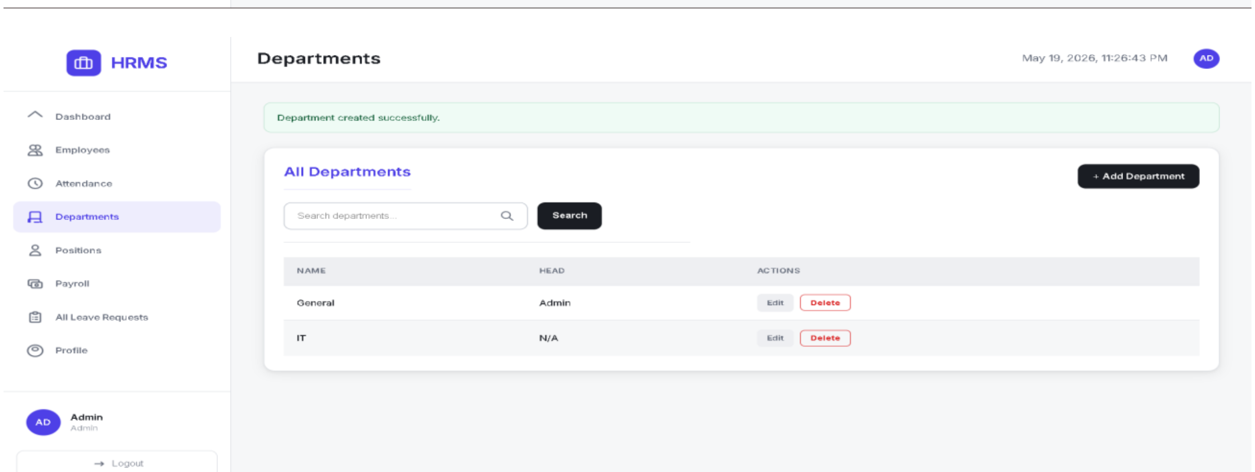
XI. Testing and Result

The system was tested through functional and user testing to ensure that all modules of the Human Resource Management System (HRMS) operated correctly and accurately. The testing included login authentication, employee management, attendance tracking, payroll computation, leave management, database validation, and user interface functionality. Different scenarios such as valid and invalid inputs, salary deductions, overtime calculations, and leave approvals were tested to verify the accuracy of the system.

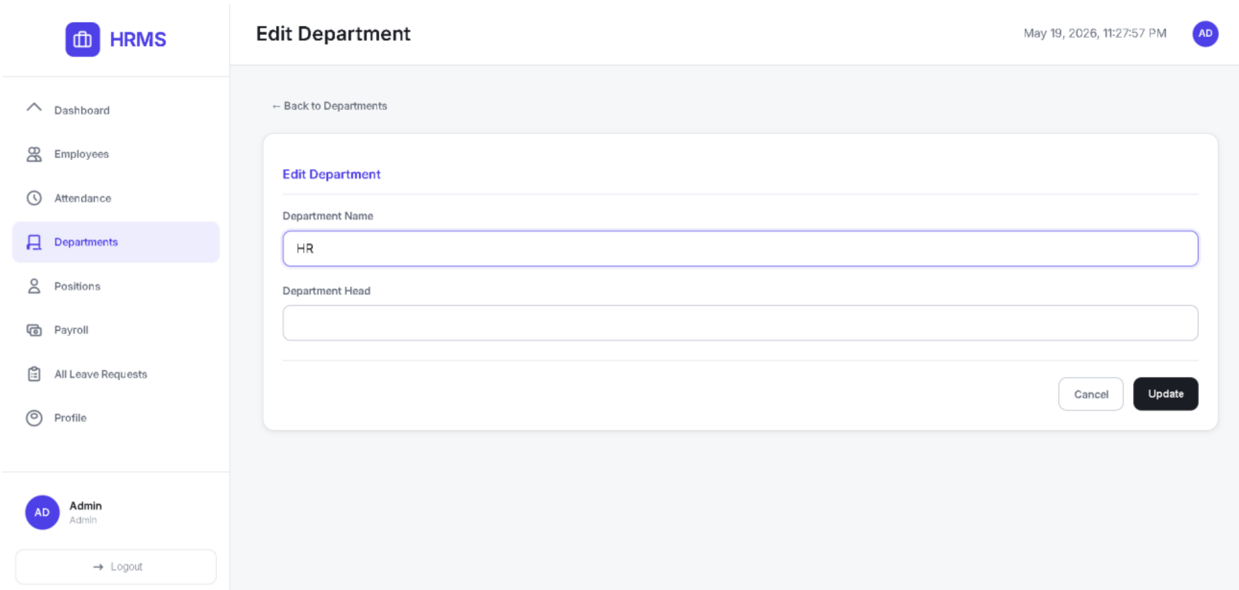
Testing (CRUD Departments)

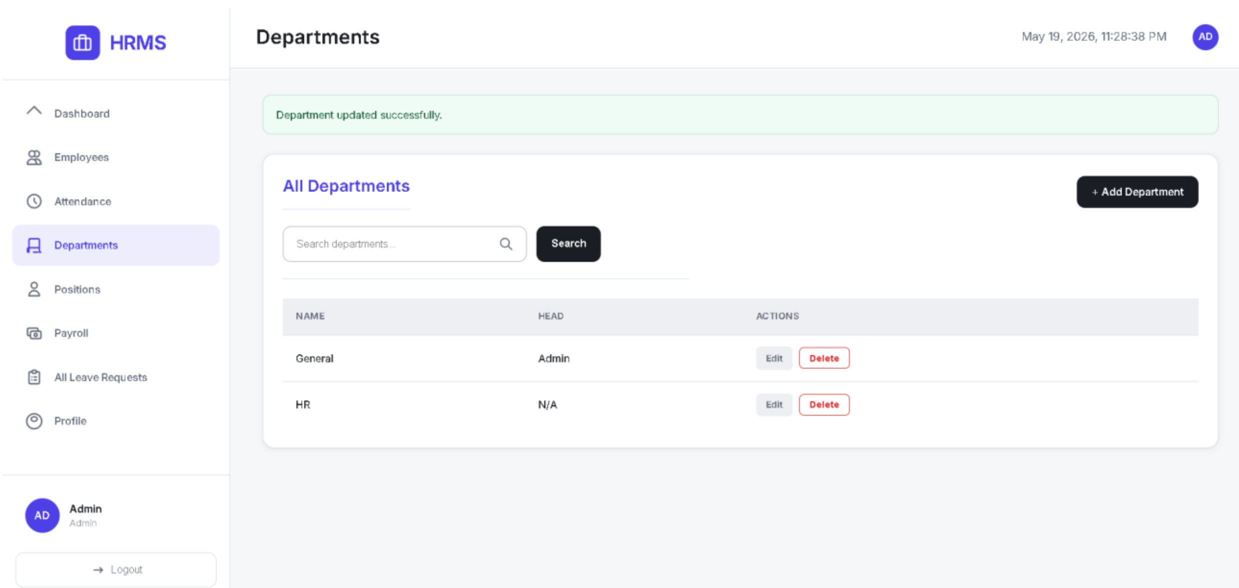
Create



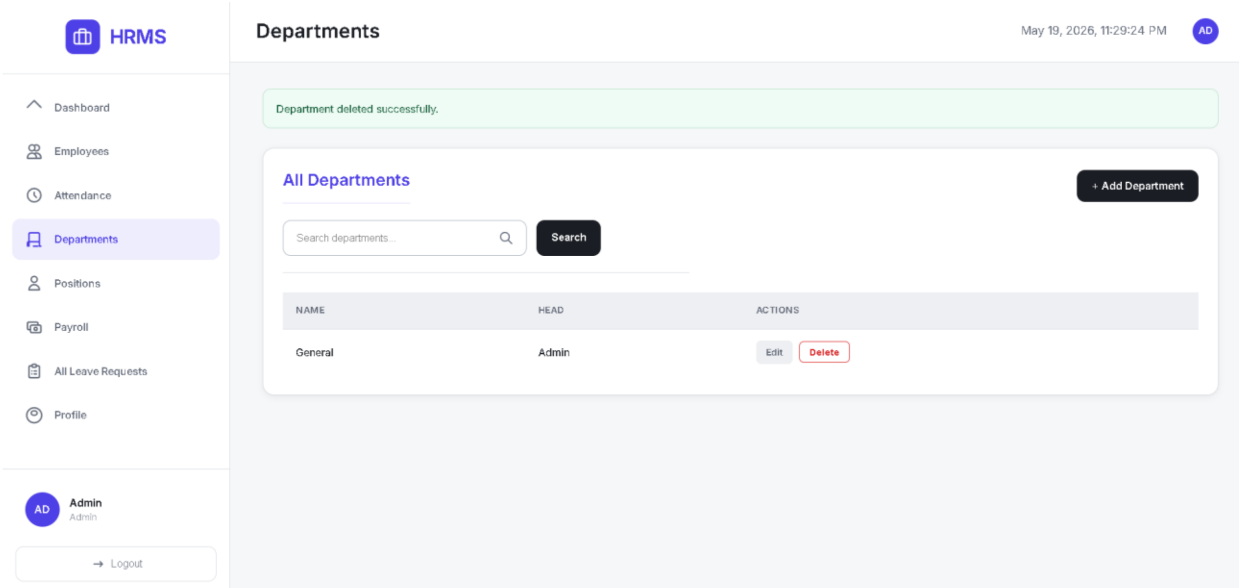


Edit / Update


HRMS
Edit Department
May 19, 2026, 11:27:57 PM
AD
Dashboard
Employees
Attendance
Departments
Positions
Payroll
All Leave Requests
Profile
← Back to Departments
Edit Department
Department Name
HR
Department Head
Cancel
Update
AD
Admin
Admin
Logout


HRMS
Departments
May 19, 2026, 11:28:38 PM
AD
Dashboard
Employees
Attendance
Departments
Positions
Payroll
All Leave Requests
Profile
Department updated successfully.
All Departments
+ Add Department
Search departments...
Search
NAME
HEAD
ACTIONS
General
Admin
Edit
Delete
HR
N/A
Edit
Delete
AD
Admin
Admin
Logout

Delete



XII. Conclusion and Recommendations

Conclusion

The Human Resource Management System (HRMS) was successfully developed to manage employee records, attendance, payroll, departments, positions, and leave management in a more efficient and organized manner. The system helps automate payroll computation by calculating salary deductions for absences and lateness, as well as additional pay for overtime. It also improves data management, reduces manual work, minimizes errors, and provides faster processing of employee information and payroll reports. Overall, the system achieved its objectives and provided a reliable solution for managing human resource operations.

Recommendations

For future improvements, the system may include additional advanced features such as biometric attendance integration, automated email notifications, employee self-service portals, and mobile accessibility. Security can also be enhanced through two-factor authentication and data encryption. Future developers may improve the payroll module by adding more detailed tax and government contribution computations based on official regulations. Generating graphical reports and analytics dashboards can also help administrators monitor employee performance and payroll expenses more effectively.